

Junchang Wang

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Education

East China Normal University Bachelor of Science 2022-
Mathematics and Applied Mathematics *Major Courses GPA: 3.95/4.0*
Courses: Object-Oriented Programming (Python, C/C++), Data Structures, Probability & Statistics
Awards: 2023-2024 National Scholarship, 2022-2023 National Scholarship

University of California, Berkeley Visiting student 2024.9 – 2025.8
Mathematics and Applied Mathematics *GPA: 3.9/4.0*
Courses: Machine Learning, Deep Learning for Visual Data, Advanced Large Language Model Agents, Great Ideas in Computer Architecture

SUMMER/WINTER SESSIONS

AI for Mathematics: Formalization and Theorem Proving Dec 2023 – Jan 2024, Peking University

Internships

NLP Intern @ HuaWei 2026.3 –
• Performed feature extraction based on open-source Omni multimodal model components, and designed downstream task models under a DLM-based generative architecture.
• Utilized the **Accelerate** and **PEFT** frameworks to conduct multi-GPU fine-tuning and other post-training procedures.
• Improved model generation speed and reduced response latency while ensuring no significant degradation in task performance metrics.

Skills

Programming Languages: Python, C, $\LaTeX$, RISC-V
Libraries: Transformers, PyTorch, PyG, Captum, NumPy, Scikit-learn, Matplotlib, Pandas
Languages: TOEFL 104

Projects

Post-training of Dream 7B with a Multi-domain CoT Dataset 2025.12 – 2026.1
• Conducted in-depth research on the Diffusion LLM architecture and its diffusion-based text generation mechanism, exploring its advantages in output diversity and controllability.
• Selected high-quality multi-domain Chain-of-Thought (CoT) datasets for supervised fine-tuning (SFT), guiding the model through structured reasoning steps.
• Performed reinforcement learning (GRPO) based on the **veRL** framework, applying CoT-optimized rewards to post-train and improve models such as Dream 7B.
• Improved accuracy on the challenging scientific reasoning benchmark GPQA, significantly enhancing performance in domain-specific question answering.

Automated Lean Theorem Proving System Based on Multi-Agent Collaboration 2025.4 – 2025.5
• Built a multi-agent collaborative architecture, improving the accuracy of code and proof generation through context separation; integrated a Lean compiler execution environment to dynamically test correctness, enabling iterative refinement and a closed-loop error feedback system.
• Enhanced reasoning quality using RAG by constructing a local knowledge base to provide contextual support, strengthening the model's capabilities in formal logical reasoning.

Implementation of Core Computer Vision Model Systems 2025.2 – 2025.5
• Designed and trained deep convolutional networks such as ResNet for high-accuracy image classification; built object detection systems based on YOLO, with a solid understanding of the multi-task learning mechanism of localization and classification.
• Implemented Vision Transformer models and applied pretrained CLIP for zero-shot image classification, gaining deep insights into multimodal representation learning and zero-shot transfer.
• Built diffusion models based on U-Net and DiT, implementing the full pipeline of image generation from noise.

Molecular marker system for pan-cancer analysis using deep learning 2022.10 – 2023.4
• Designed and implemented a high-accuracy classification model, using SMOTE for data augmentation to handle high-throughput genomics data.
• Developed Graph Neural Networks based on GraphSAGE and Autoencoder architectures for feature extraction and dimensionality reduction.
• Applied Explainable AI techniques (Integrated Gradients, GNNExplainer) to identify key gene biomarkers and interpret their biological significance.